

An API for procedural generation of curvilinear interface components

Luke Ker

Submitted in partial fulfilment of the requirements of
Edinburgh Napier University
for the Degree of
MSc Computing

School of Computing

August 2026

Authorship Declaration

I, *Luke Ker*, confirm that this dissertation and the work presented in it and my own achievement.

1. Where I have consulted the published work of others this is always clearly attributed;
2. Where I have quoted from the work of others the source is always given. With the exception of such quotations this dissertation is entirely my own work;
3. I have acknowledged all main sources of help;
4. If my research follows on from previous work or is part of any larger collaborative research project I have made clear exactly what was done by others and what I have contributed myself;
5. I have read and understood the penalties associated with Academic Misconduct.
6. I also confirm that I have obtained **informed consent** from all people I have involved in the work in this dissertation following the School's ethical guidelines.

Type name: *Luke Ker*

Date: 24th July 2026

Matriculation No: 40583024

General Data Protection Regulation Declaration

Under the General Data Protection Regulation (GDPR) (EU) 2016/679, the University cannot disclose your grade to an unauthorised person. However, other students benefit from studying dissertations that have their grades attached.

Please sign your name under *one* of the options below to state your preference.

The University may make this dissertation, with indicative grade, available to others.

Luke Ker

The University may make this dissertation available to others, but the grade may not be disclosed.

The University may not make this dissertation available to others.

Abstract

Insert abstract here, no more than 300 words in length.

Contents

1	Introduction	1
1.1	Background	1
1.2	Problem Statements	1
1.3	Aims and Objectives	1
1.4	Research Questions	1
1.5	Project Scope	1
1.6	Document Structure	1
2	Literature and Technology Review	4
2.1	Introduction	4
2.2	Topic 1	4
2.3	Summary	4
3	Design	5
3.1	Introduction	5
3.2	Design	5
3.2.1	Sub-design	6
3.3	Conclusions	6
4	Implementation	7
4.1	Intoduction	7
4.2	Conclusions	7
5	Evaluation	8
5.1	Introduction	8
5.2	Conclusions	8
6	Conclusions	9
6.1	Main Conclusions	9
6.2	Delivery against objectives	9
6.3	Future work	10

6.4	Personal reflection	10
-----	-------------------------------	----

List of Tables

3.1	My table	6
-----	--------------------	---

List of Figures

1.1	My figure	2
-----	---------------------	---

Acknowledgements

It is normal to thank those who have given help and support (typically your supervisor should get a kind mention!). Keep acknowledgements short and business-like, but do take care to remember all those who deserve credit. This is especially important where outside agencies are involved, or where experimental work has been carried out. Remember to include the names of proof-readers.

Chapter 1

Introduction

1.1 Background

This is a citation (Van Velthoven and Cordon, 2019). Citations should be made using the `natbib` package, using the `citep` and `citet` commands.

1.2 Problem Statements

1.3 Aims and Objectives

Blah as shown in Figure 1.1. And with User Interface (UI), which is abbreviated as UI.

- Blah
- Blah

1.4 Research Questions

Blah

- Blah

1.5 Project Scope

1.6 Document Structure

Blah

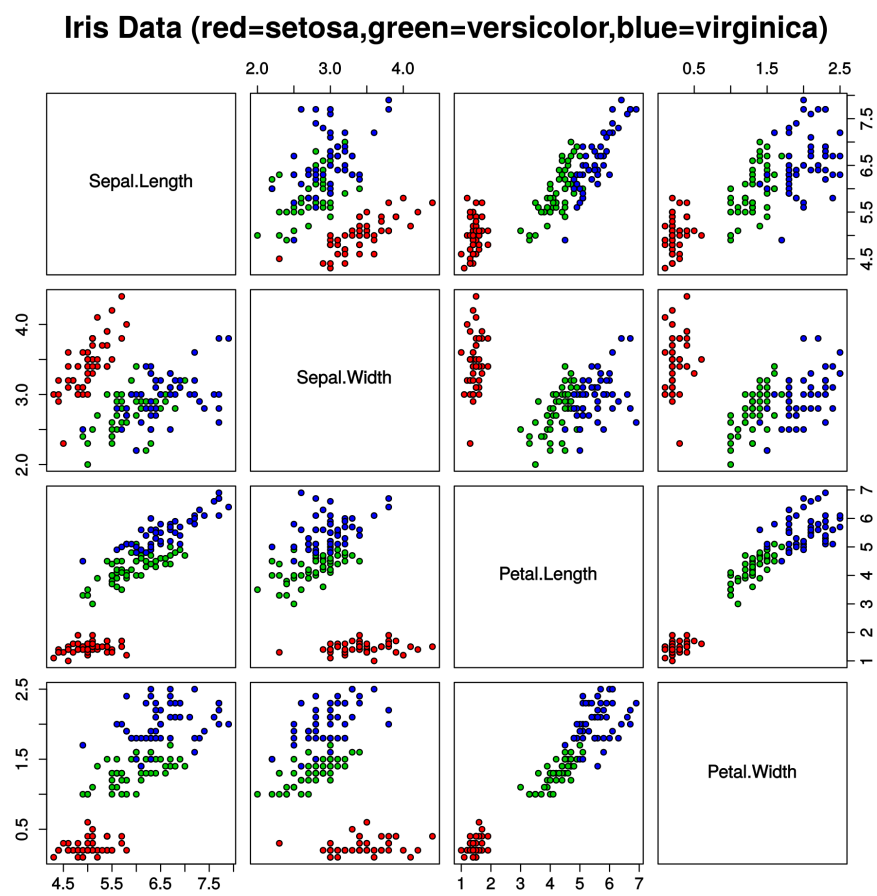


Figure 1.1: My figure

- Blah
- Blah

Chapter 2

Literature and Technology Review

2.1 Introduction

2.2 Topic 1

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

2.3 Summary

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

Chapter 3

Design

3.1 Introduction

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

3.2 Design

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere. In Table 3.2.

Security features	Handling Security Service	References
Key 1	Auth 1	5, 5
Key 2	Auth 2	7, 5
Key 3	Auth 3	4, 5

Table 3.1: My table

3.2.1 Sub-design

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, duis nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

3.3 Conclusions

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, duis nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

Chapter 4

Implementation

4.1 Introduction

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

4.2 Conclusions

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

Chapter 5

Evaluation

5.1 Introduction

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

5.2 Conclusions

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

Chapter 6

Conclusions

6.1 Main Conclusions

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

6.2 Delivery against objectives

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

6.3 Future work

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

6.4 Personal reflection

Lorem ipsum dolor sit amet, faucibus lorem id, nec vel curabitur fusce feugiat. Cupiditate sollicitudin delectus turpis at tincidunt ut, lorem volutpat praesent massa vivamus justo, semper magna, ipsum eros gravida vestibulum dolor lorem expedita. Tellus ac pretium feugiat massa, praesent tincidunt curabitur gravida lorem sed nisl, eu hac sit eaque pede aenean, suspendisse aenean tincidunt suspendisse ut scelerisque, fringilla magna natoque ipsum ac. Eget mattis. Nunc donec magna augue, venenatis proin commodo non feugiat mauris, mollis sodales accumsan dignissim eget, suscipit nisl imperdiet nibh integer, dui nunc dolor hac. Ac sem. Nulla ut fames dolorem, donec at tortor posuere.

References

Van Velthoven, M. H. and Cordon, C. (2019). Sustainable adoption of digital health innovations: Perspectives from a stakeholder workshop. *Journal of medical Internet research*, 21(3):e11922.

REFERENCES
